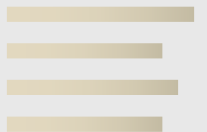


İstatistik ve



OlASILIK



1 ödev %20 → Bize verilen soruları çözüp yollayacağız
 1 vize %40
 1 final %40

Sets (Kümeler)

Each element should be unique

Subsets (Alt Kümeler)

Super Set [Alt Kümelerin Tasi]

Set Operations

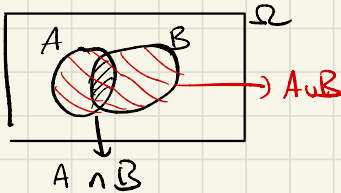
* Union (Birleşim)

$$x \in (A \cup B) \iff (x \in A) \text{ or } (x \in B)$$

$$\begin{matrix} \downarrow & & \downarrow \\ \{1, 3\} & & \{2, 4\} \end{matrix}$$

$$A \cup B \rightarrow \{1, 2, 4\}$$

* Intersection (Kesişim)

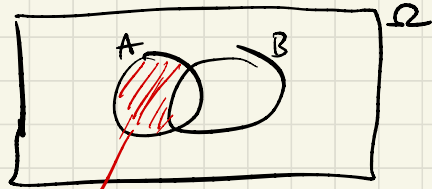


* Partition Sets

a) They are disjoint

$$b) A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n = \Omega, S$$

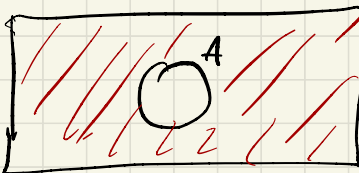
* Difference (Subtraction) [Fark]



$$A - B \equiv x \in A \text{ and } x \notin B$$

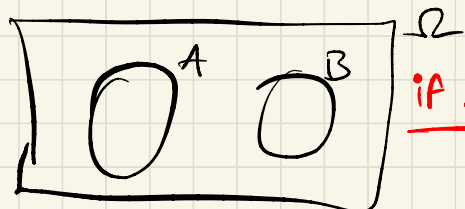
$$A/B, A \cap B'$$

* Complement (Tümleyen)



$$A', \bar{A}, A^c$$

* Disjoint Sets (Ayrık Kümeler)



$$\text{if } A \cap B = \emptyset$$

* De Morgan's Law

$$(A \cup B)' \equiv \overline{A \cup B} \equiv A' \cap B'$$

Ex $S = \{1, 2, 3, 4, 5, 6\}$ $A = \{1, 2\}$ $B = \{2, 4, 6\}$

$$A \cup B = \{1, 2, 4, 6\} \quad (A \cup B)' = \{3, 5\}$$

$$A \cap B = \{2\}$$

$$A' = \{3, 4, 5, 6\}$$

$$B' = \{1, 3, 5\}$$

* Distributive Law

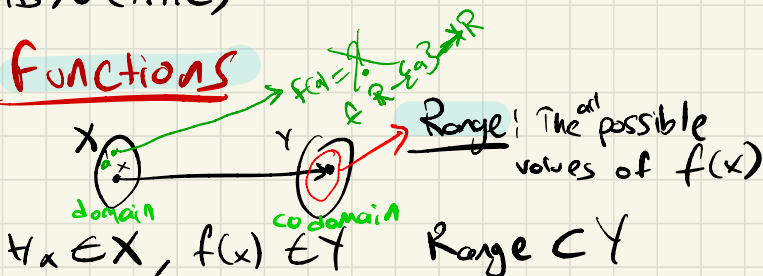
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Functions

$$f: X \rightarrow Y$$

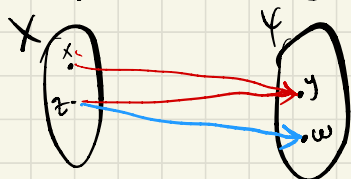
domain co-domain



$$f: \mathbb{R} \rightarrow \mathbb{R} \text{ defined as } f(x) = x^2$$

$$X = Y = \mathbb{R}$$

$$\text{Range}(f) = \mathbb{R}^+ = \{x \in \mathbb{R} \mid x \geq 0\}$$



$$f(x_1) = y \quad \checkmark$$

$$f(z) = y \quad \checkmark$$

$$f(z) = y$$

$$f(z) = w \quad \times$$

This one can happen

This one can not happen

Countable and Uncountable Sets

- A set is finite if $|A| < \infty$

- A set is countable if it is finite or its elements can be enumerated or listed in a sequence

$|A| \rightarrow$ cardinality shows the number of elements.

$$A = \bigcup_{k=1}^{\infty} \{a_k\} \rightarrow A = \{a_1, a_2, a_3, \dots\}$$

- Finite Sets

$$A = \{1, 2, 4, 8, 9\}$$

$$x \in [4, 8) \text{ s.t. } x \in \mathbb{Z}$$

$\downarrow$
4, 5, 6, 7

Infinite Set:

$$\mathbb{Z}^+ = \{1, 2, 3, 4, \dots\}$$

Uncountable Sets

The elements can not be enumerated.

$\mathbb{R} \rightarrow$ real numbers No Enumerated

A set is countable infinite if it is in one-to-one correspondence with

$$\mathbb{N} = \{1, 2, 3, \dots\} = \bigcup_{k=1}^{\infty} \{k\}$$

$$\mathbb{Q}^+ = \left\{ \frac{a}{b} : \begin{array}{l} \text{at least} \\ a \text{ or } b \end{array} > 0 \right\} \rightarrow \begin{array}{l} \text{countable} \\ \text{infinite} \end{array}$$

$$\mathbb{Z} = \{\dots, 0, 1, 2, \dots\} \rightarrow \text{countably infinite}$$

Power Sets

Sets of all subsets of a set A

$$A = \{1, 2\}$$

$$P(A) = \{\{1\}, \{2\}, \{1, 2\}, \{\}\}$$

Exhaustive Sets — The sets shouldn't be disjoint

Sets A, B, C, D are exhaustive if their union is Ω

$$A \cup B \cup C \cup D = \Omega$$

PROBABILITY THEORY

Random experiment: A phenomenon whose outcome can not be predicted with certainty

- Rolling a die
- Flipping a coin

→ Event is a set

→ Sample space is a Universal Set

Outcome → Element of a set

Event = Collection of possible outcomes

Sample Space = Set of all possible outcomes

- Roll a die 3 times

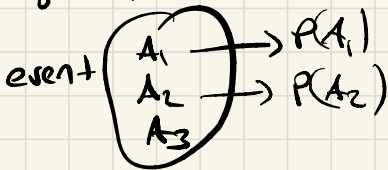
$$\Omega = \{(1,1,1), (1,1,2), (1,1,3), \dots, (6,6,6)\}$$

216 possible outcome

Probability

An event A and $P(A)$: Probability of A

We assign probability $P(A)$ to every event A .



Axioms of Probability: $P(\cdot)$ is a function that maps events in the sample space to real numbers

① - For any event A , $P(A) \geq 0$.

② - Probability of the sample space Ω $P(\Omega) = 1$

③ For any countable collection $A_1, A_2, \dots$ of disjoint events

$$P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + P(A_3) + \dots$$

2. hafta Bitimi

3. hafta Başlangıcı

Probability Rules

$$\textcircled{1} P(A) = 1 - P(\bar{A}) \quad 1 = P(\Omega)$$

$$\textcircled{2} P(\emptyset) = 0 \quad 0 = P(\bar{\Omega})$$

$$\textcircled{3} 0 \leq P(A) \leq 1$$

$$\textcircled{4} P(A-B) = P(A) - P(A \cap B)$$

$$\textcircled{5} P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\textcircled{6} A \subset B \Rightarrow P(A) \leq P(B)$$

Axioms are important

↳ Rules of prob. are obtained using these axioms

↳ Any function that obeys these axioms define a prob. measure

Calculating Prob.

• Equally likely outcomes:

> Rolling a fair die

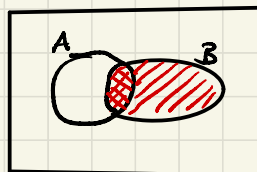
$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

$$P(1) = P(2) = P(3) = P(4) = P(5) = P(6) \Rightarrow P(\Omega) = 1$$

Conditional Probability (Kosullu)

Definition: The conditional Probability $P(A|B)$ prob. that A occurred given B has occurred is

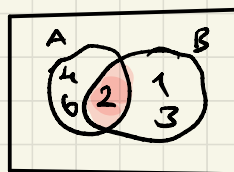
$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$



Ex: Roll a die. Prob. of getting an even number given the outcome is less than or equal to 3?

$$P(\text{even} | \underbrace{\text{outcome} \leq 3}_B)$$

$\uparrow$
 $P(A)$



$$P(A) = \{2, 4, 6\}$$

$$(A \cap B) = \{2\}$$

$$P(B) = \{1, 2, 3\}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1}{3} \rightarrow \begin{matrix} \nearrow |A \cap B| \\ \searrow |B| \end{matrix}$$

Ex: We know that 90% of the flights depart on time
80% of the flights arrive on time
70% of the flights both arrive and depart on time

a) We know that flight departed on time, Prob. that it will arrive on time?

90% depart
 A

80% arrive
 B

70% both arrive and depart
 $P(A \cap B)$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{70}{90}$$

If there are more than 2 events:

$P(\text{rain, clouds, windy})$

Chain Rule = $P(A \cap B) = P(A|B) \cdot P(B)$

$$P(A \cap B \cap C) = \underbrace{P(C) \cdot P(B|C)}_{P(B \cap C)} \cdot P(A|BC)$$

$$P(A|B, C) = \frac{P(A \cap B \cap C)}{P(B \cap C)}$$

$$\rightarrow P(A|B) = 1 - P(\bar{A}|B)$$

Independence of events

Definition: 2 events A and B are independent iff: $\Leftrightarrow$

$$P(A|B) = P(A)$$

Independents events
are not disjoint.

$$P(A \cap B) = P(A) \cdot P(B)$$

Ex: I roll a die and observe number X. Let A be an event that X is an even number and let B an event that $X > 4$

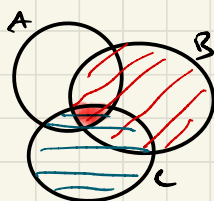
Are A and B is independent?

$$A = \{2, 4, 6\} \rightarrow P(A) = \frac{1}{2}$$

$$B = \{5, 6\} \rightarrow P(B) = \frac{1}{3}$$

$$P(A \cap B) = \frac{1}{6} = \frac{1}{2} \cdot \frac{1}{3} = P(A) \cdot P(B)$$

A and B are Independent.



Conditional Independence

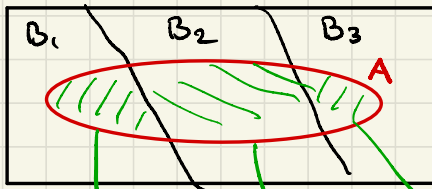
$$P(A \cap B) = P(A) \cdot P(B)$$

Two events A and B are conditionally independent given C iff

$$P(A \cap B | C) = P(A | C) \cdot P(B | C)$$

2 Law of Total Probability

Let $B_1, B_2, B_3, \dots, B_n$ be a partition of Ω and $P(B_i) > 0$ for any event A we have



$$P(A) = P(A \cap B_1) + P(A \cap B_2) + P(A \cap B_3)$$

Ex:



3 unfair coins

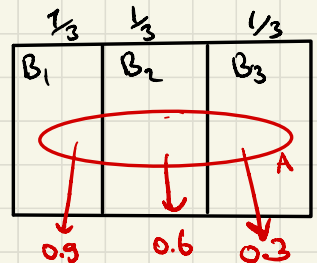
We know that:

- Coin 1 $\rightarrow P(H|C_1) = 0.9 - P(T) = 0.1$
- Coin 2 $\rightarrow P(H|C_2) = 0.6$
- Coin 3 $\rightarrow P(H|C_3) = 0.3 - P(T) = 0.7$

We draw a coin randomly and we flip

$$P(H) = ? \quad P(A) = \sum_i P(A \cap B_i) = \sum_i P(A | B_i) \cdot P(B_i)$$

$$= 0.9 \cdot \frac{1}{3} + \frac{0.6}{3} + \frac{0.3}{3} = \underline{\underline{0.6}}$$



★ Bayes Rule ★

$$P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$

Ex: In Previous example,

If a draw Heads, prob. of drawing coin 1?

$$\left. \begin{array}{l} P(H|C_1) = 0.9 \\ P(H|C_2) = 0.6 \\ P(H|C_3) = 0.3 \end{array} \right\} \begin{array}{l} \text{Bayes} \\ \text{Rule} \end{array} \quad P(C_1|H) = \frac{P(H|C_1) \cdot P(C_1)}{P(H)}$$

$$P(C_1) = P(C_2) = P(C_3) = \frac{1}{3}$$

$$\frac{0.9 \cdot \frac{1}{3}}{0.6} = \frac{1}{2}$$

3. hafta Bitimi

4. hafta Başlangıcı

Counting: $P(A) = \frac{|A|}{|\Omega|}$ } equally likely outcome

The number of elements $|\Omega|$

Multiplication Principle:

Coffe

1 3 type
2 3 size $> 3 \times 3 = 9$ possibilities

Ex: Choosing a Password:

- begin with 2 lower case
- 1 Capital letter forward by.
- 4 digits

l l C d d d d

$$26 \times 26 \times 26 \times 10 \times 10 \times 10 \times 10 = 26^3 \times 10^4 \rightarrow$$

Sampling:

Sampling from a set: Randomly choosing an element from a set.

With Replacement: Put the sampled element back. →

Without Replacement: Repetition is not allowed.

Do not put the sampled obj. back

Ordering = $(a_1, a_2, a_3) = (a_1, a_2, a_3)$

Unordered Sampling: $(a_1, a_2, a_3) = (a_3, a_2, a_1)$

SAMPLING



ORDERED Sampling with Replacement =

ex: $A = \{1, 2, 3\}$ draw $k=2$ samples. In how many ways the sampling can be done?

it can be cause of ordered sampling

- $(1, 2)$
- $(1, 3)$
- $(1, 1) \rightarrow$ this can be cause of with Rep.
- $(2, 1) \rightarrow$

different ways

$(2, 2)$

$(2, 3)$

$(3, 1)$

$(3, 2)$

$(3, 3)$

$\rightarrow$ What is the number of possible lists?

$$\frac{n^k}{=}$$

$\rightarrow$ Number of subsets?

$$2^n$$

ORDERED Sampling Without Replacement =

- No repetition

ex: $A = \{1, 2, 3\}$ $k=2$

$\rightarrow (1, 1)$ $\times \rightarrow$ Without Replacement!

$\rightarrow (1, 2)$ $\checkmark$

$\rightarrow (1, 3)$ $\checkmark$

$\rightarrow (2, 1)$ $\checkmark$

$\rightarrow (2, 2)$ $\times \rightarrow$ w.o.R

$\rightarrow (2, 3)$ $\checkmark$

$\rightarrow (3, 1)$ $\checkmark$

$\rightarrow (3, 2)$ $\checkmark$

$\rightarrow (3, 3)$ $\times \rightarrow$ w.o.R

6 possible ways $\rightarrow$

$$\frac{n!}{(n-k)!} \left. \vphantom{\frac{n!}{(n-k)!}} \right\} \begin{matrix} k \\ \text{Permutation} \end{matrix}$$

ex: K people in a class. What is the probability that at least two of them have the same birthday?

$n \rightarrow 365$ days

A = event that 2 people have same b-day

A^c = No two people have same b-day

$$p(A) = 1 - \frac{(A^c)}{(2)} \rightarrow \frac{1}{365} \cdot \frac{2}{364} \cdot \frac{3}{363} \cdots \frac{k}{365-k+1} = \frac{n!}{(n-k)!}$$

$$\downarrow$$
$$(365)^k$$

$$p(A) = 1 - \frac{\frac{n!}{(n-k)!}}{n^k}$$

Unordered Sampling without Replacement =

- No ordering and no replacement.

ex: $A = \{1, 2, 3\}$ $k=2$?

(1,1) X $\rightarrow$ w.o.R

(1,2) ✓

(1,3) ✓

(2,1) X $\rightarrow$ Unordered

(2,2) X $\rightarrow$ w.o.R

(2,3) ✓

(3,1) X $\rightarrow$ Unordered

(3,2) X $\rightarrow$ Unordered

(3,3) X $\rightarrow$ w.o.R

3 ways

k combination

$$\binom{n}{k} \rightarrow \frac{n!}{k!(n-k)!} \rightarrow \frac{3 \cdot 2 \cdot 1}{2! \cdot 1!} = 3$$

Ex = Choose 3 cards from a deck (52 cards). What is the probability that these cards contain at least one Ace? 4-Ace in a deck

$$P(A^c) = \text{None of 3 cards contain one Ace} = \binom{52-4}{3}$$

$$P(Q) = \binom{52}{3} \quad P(A) = 1 - \frac{\binom{48}{3}}{\binom{52}{3}} \rightarrow$$

$$\frac{12}{12} \quad \frac{11}{12} \quad \frac{10}{12} \quad \frac{9}{12}$$

Binomial Coefficients

$\binom{n}{k}$ is called Binomial Coefficients

$$(a+b)^n = n a^{n-1} b + ? a^2 b^2 + ? a^3 b^3 \dots$$

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} \rightarrow (a+b)^2 = \binom{2}{0} a^0 b^2 + \binom{2}{1} a b + \binom{2}{2} a^2 b^0$$

$$\text{Binomial formula} = \binom{n}{k} p^k (1-p)^{n-k}$$

ex = 10 people went to restaurant.

5 main dish

3 drinks

2 dessert

In how many ways can these people be selected?

$$\binom{10}{5} \binom{5}{3} \binom{2}{2} = \frac{10!}{5!5!} \times \frac{5!}{3!2!} \times \frac{2!}{2!0!} = 1$$

Unordered Sampling With Replacement =

Ex = $A = \{1, 2, 3\}$

$(1, 1)$ ✓

$(1, 2)$ ✓

$(1, 3)$ ✓

$(2, 1)$ ✗

$(2, 2)$ ✓

$(2, 3)$ ✓

$(3, 1)$ ✗

$(3, 2)$ ✗

$(3, 3)$ ✓

6 possible ways

$$\binom{n+k-1}{k} \xrightarrow{\text{with Rep.}} \binom{n+k-1}{n-1}$$

ex = 10 passengers get on airport shuttle. The shuttle route includes 5 hotels.

The driver records how many passengers leave shuttle at each hotel. How many possible lists?

x_1	x_2	x_3	x_4	x_5
10	0	0	0	0
9	1	0	0	0
9	0	1	0	0
}				

$$\binom{5+10-1}{10} = \binom{5+10-1}{4} = \frac{15 \times 14 \times 13 \times 12}{24}$$

$$77.13 = 901$$

Random Variables (function)

Random variable is a real valued function that assigns numerical values to possible outcomes of the random experiments

Ex: Toss a coin 3 times X is the number of heads

Range of $X =$

$$\text{Range}(X) = R_X = \{0, 1, 2, 3\}$$

H H H
H H T
H T T
T T T

Ex: if the Ranges are countable, X is a discrete random variable.
uncountable $\rightarrow$ continuous random variable

Probability Mass Function

Let X be a random variable with range $R_X = \{x_1, x_2, \dots\}$ (finite, countably infinite)

The function

$$P_X(x_k) = P(X = x_k) \text{ for } k = 1, 2, 3, \dots$$

Ex: Toss a coin twice, X is the number of heads

- Range of X $R_X = \{0, 1, 2\}$

- P_X

H H	2
T H	1
H T	1
T T	0

$$\begin{aligned} P_X(0) &= P(X=0) = \frac{1}{4} \\ P_X(1) &= P(X=1) \rightarrow \frac{2}{4} \\ P_X(2) &= P(X=2) \rightarrow \frac{1}{4} \end{aligned}$$

EX: We have unfair coin $P(H) = p$ & $p < 1$

I toss the coin until I get a H. Y is the total number of tosses. Distribution of Y ?

$$R_Y = N = \{1, 2, 3, \dots\}$$

$$P_Y(1) = (Y=1) = p$$

$$(1-p) \text{ TH} \leftarrow P_Y(2) = (Y=2) = (1-p) \cdot p$$

$$\text{TTTH} \leftarrow P_Y(3) = (Y=3) = (1-p)(1-p) \cdot p$$

$$k^{\text{th}} \rightarrow H \quad P_Y(k) = (Y=k) = (1-p)^{k-1} \cdot p \quad \leftarrow$$

$\underbrace{(k-1)}_{\text{T...T}}$
 $k-1$

$$P_Y(k) = \begin{cases} (1-p)^{k-1} p & \text{for } k=1, 2, 3, \dots \\ 0 & \text{otherwise} \end{cases}$$

PMF is a probability measure

$$> 1 - 0 \leq P_X(x) \leq 1 \quad \text{for all } x$$

$$> 2 - \sum_{x \in R_X} P_X(x) = 1$$

$$> 3 - \text{for any set } A \subset R_X \quad P(X \in A) = \sum_{x \in A} P_X(x)$$

EX: Let X be discrete random variable

$$R_X = \{1, 2, 3, 4\}. \quad P_X(1) = P_X(2) = \frac{1}{3} \quad \text{and} \quad P_X(4) = \frac{1}{6}$$

$$\rightarrow \text{a) } P_X(3) = ? \rightarrow \sum_{x \in R_X} P_X(x) = 1 \Rightarrow \left\{ \frac{1}{3} + \frac{1}{3} + \underline{P_X(3)} + \frac{1}{6} = 1 \right\} \rightarrow P_X(3) = \frac{1}{6} //$$

$$\rightarrow \text{b) } P(2 \leq X < 4) = ? \\ = P(X \in \{2, 3\}) = P_X(2) + P_X(3) = \frac{1}{3} + \frac{1}{6} = \frac{1}{2} //$$

Independent Random Variables

Ex: Roll a die twice and get two numbers X and Y .

a) Find R_X, R_Y and PMF of X and Y .

$$R_X = R_Y = \{1, 2, 3, 4, 5, 6\}$$

$$P_X(1) = \frac{1}{6} = P_X(2) = P_X(3) = P_X(4) = P_X(5) = P_X(6)$$

$$P_Y(1) = \frac{1}{6} = P_Y(2) = P_Y(3) = P_Y(4) = P_Y(5) = P_Y(6)$$

b) Find $P(X=2 \text{ and } Y=6) = P_X(2) \cdot P_Y(6) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$

c) Find $P(\underbrace{X > 3}_{\{4, 5, 6\}} \text{ and } Y=2) = \underbrace{[P_X(4) + P_X(5) + P_X(6)]}_{= \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}} \cdot P_Y(2)$

d) Let $Z = X + Y$ Range of Z ?

$$Z = X + Y$$

$$R_X = R_Y = \{1, 2, 3, 4, 5, 6\}$$

$$P_Z =$$

$$P_X(1) = P_Y(1) = P_X(2) = \dots \quad P_X(6) = \frac{1}{6}$$

$$P_Z(k) = \begin{cases} \frac{1}{6} & k = \{1, 2, \dots, 6\} \\ 0 & \text{otherwise} \end{cases}$$

$$P_Z(2) = P(X + Y = 2)$$

$$= P(X=1, Y=1) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

$$P_Z(3) = P(X + Y = 3) = \underbrace{P(X=1, Y=2)}_{= P(X=1) \cdot P(Y=2)} + \underbrace{P(X=2, Y=1)}_{= P(X=2) \cdot P(Y=1)}$$

$$P_Z(4) = P(X + Y = 4) = \underbrace{P(X=1, Y=3)}_{= P(X=1) \cdot P(Y=3)} + \underbrace{P(X=2, Y=2)}_{= P(X=2) \cdot P(Y=2)} + \underbrace{P(X=3, Y=1)}_{= P(X=3) \cdot P(Y=1)}$$

Ex: I toss a coin twice.

X = number of heads I observe

→ I toss the coin two more times.

Y = number of heads I observe

$$P(X < 2 \text{ and } Y > 1) = ?$$

sol: $P(X < 2 \text{ and } Y > 1) = P(X < 2) \cdot P(Y > 1)$

$$= [P_X(0) + P_X(1)] \cdot P_Y(2)$$

$$= \left[\frac{1}{4} + \frac{1}{2} \right] \cdot \frac{1}{4} = \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{16} //$$

$$\begin{aligned} (2) & \rightarrow HH \rightarrow \frac{1}{4} \\ 1 & \left\{ \begin{array}{l} HT \\ TH \end{array} \right\} \frac{1}{2} \\ 0 & \left\{ \begin{array}{l} TT \end{array} \right\} \frac{1}{4} \\ & \quad \quad \quad 4 \end{aligned}$$

Special distributions

1. Bernoulli Distribution

→ A Bernoulli trial is a random experiment that has two possible outcomes which can be labeled as success or failure

→ Simplest discrete random variable

$$P_X(x) = \begin{cases} 1 & \text{for } x=1 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{Child} \begin{cases} G & 1 \\ B & 0 \end{cases}$$

$$I_A = \begin{cases} 1 & \text{success} \\ 0 & \text{failure} \end{cases} \quad I_A \sim \text{Bernoulli}(P(A))$$

$$\text{Coin} \begin{cases} 1 & \text{Head} \\ 0 & \text{Tail} \end{cases}$$

Definition: A random variable X is said to be a Bernoulli Random Variable with parameter p , $X \sim \text{Bernoulli}(p)$ if its PMF

$$P_X(x) = \begin{cases} p & \text{for } x=1 \\ 1-p & \text{for } x=0 \\ 0 & \text{otherwise} \end{cases}$$

Indicator random variable:
Bernoulli r.v (BRV) is also called indicator R.V.

2. Geometric Distribution

→ Suppose we have a coin.

$P(H)=p$. I toss the coin until I observe the first head.
We define X as the number coin tosses. X is said to have geometric distribution.

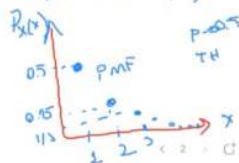
- Repeating Bernoulli trials until the first success.

$$\underbrace{T \dots T}_{(k-1) \text{ tosses}} \underbrace{H}_{(1-p)} \rightarrow k \text{ tosses} \quad \left. \begin{matrix} p \cdot (1-p)^{k-1} \end{matrix} \right\} \text{ for } k=1, 2, 3, \dots$$

Definition

A random variable X is said to be a geometric random variable with parameter p / $X \sim \text{Geometric}(p)$, its PMF is

$$P_X(k) = \begin{cases} p \cdot (1-p)^{k-1} & \text{for } k=1, 2, \dots \\ 0 & \text{otherwise.} \end{cases}$$



3. Binomial Distribution

→ $P(H)=p$ Toss a coin n times.

X = number of heads

$$R_X = \{0, 1, \dots, n\}$$

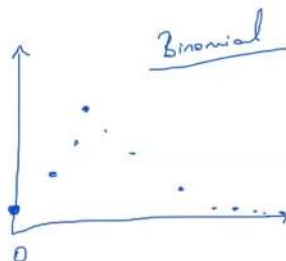
→ n independent Bernoulli trials

PMF of X

$$\binom{n}{k} p^k (1-p)^{n-k}$$

Df: A random variable X is said to be binomial random variable with parameters n and p , shown as $X \sim \text{Binomial}(n, p)$ and its PMF

$$P_X(k) = \begin{cases} \binom{n}{k} p^k (1-p)^{n-k} & \text{for } k=0, 1, 2, \dots, n \\ 0 & \text{otherwise} \end{cases}$$



Binomial is sum of Bernoullies

$X_1, X_2, \dots, X_n$ are independent Bernoulli(p) random variables

$X = X_1 + X_2 + \dots + X_n$ has a Binomial (n, p) distribution.

Ex: $X \sim \text{Binomial}(n, p)$ and $Y \sim \text{Binomial}(m, p)$

$Z = X + Y$. Find PMF of Z .

$$\left. \begin{array}{l} X = X_1 + X_2 + \dots + X_n \\ Y = Y_1 + Y_2 + \dots + Y_m \end{array} \right\} n+m$$

$Z = X + Y = X_1 + X_2 + \dots + X_n + Y_1 + \dots + Y_m$ since X_i and Y_j are independent Bernoulli trials.

$$P_Z(k) = \begin{cases} \binom{n+m}{k} p^k \underbrace{(1-p)^{n+m-k}}_{\text{tails}} & \text{for } k=0, 1, \dots, n+m \\ 0 & \text{otherwise} \end{cases}$$

4. Pascal (Negative Binomial) Distribution

- It is the generalization of geometric distribution.
- Random exp. of independent trials until observing m successes.

- Coin $P(H) = p$
- Toss the coin until I observe m Heads.

- X as the number of coin tosses.

X is said to have Pascal distribution with parameters (p, m)

$$X \sim \text{Pascal}(m, p)$$

If $m=1$ Pascal($1, p$) = Geometric

$$\text{Range of } X \quad R_X = \{m, m+1, m+2, \dots\}$$

$$\text{event } A = \{X=k\}$$

$A = B \cap C$ where

B event that we have $m-1$ heads in first $k-1$ trials.

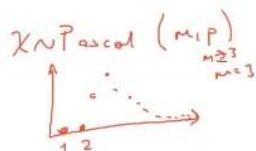
C event that we observe a head in k^{th} trial.

$$P(A) = P(B \cap C) = \underbrace{P(B)}_{\text{Binomial}} \cdot \underbrace{P(C)}_{p}$$

$$P(A) = P(B) \cdot P(C)$$

$$= \binom{k-1}{m-1} p^{m-1} (1-p)^{k-m} \times p$$

$$= \binom{k-1}{m-1} p^m (1-p)^{k-m}$$



$$k - (k-1) = 1$$

$$P(B) = \binom{k-1}{m-1} p^{m-1} (1-p)$$

$$= \binom{k-1}{m-1} p^{m-1} (1-p)^{k-m}$$

5. Poisson Distribution

- Used in cases when we are counting the occurrences of certain events in an interval of time or space.

→ Approximation of real-world random variable.

→ We are counting the number of customers at a certain time period.
On average $\lambda = 15$ customers visit the store.

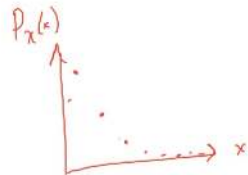


$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$p = \frac{\lambda}{n} \rightarrow n$ is very large

$$\lim_{n \rightarrow \infty} P(X=k) = \frac{\text{solve}}{\text{Books! check}} = \frac{e^{-\lambda} \lambda^k}{k!} \text{ for } k \in \mathbb{R}_x$$

Poisson dist.



EX: Number of emails I get model Poisson distribution

→ Avg. emails per minute 0.2

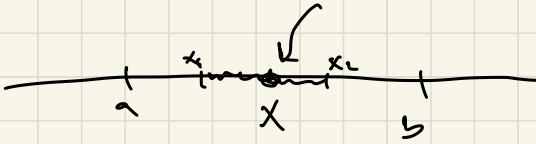
- What is the probability that I get no emails in an interval of 5 minutes?

(a) λ is a Poisson random variable $\lambda = 5 \cdot 0.2 = 1$

$$P(X=0) = P_{\lambda}(0) = \frac{e^{-\lambda} \lambda^0}{0!} = \frac{e^{-1} \cdot 1}{1} = \frac{1}{e} \approx \frac{1}{2.71}$$

(b) 5 dk. er af 3 email? → 1 - (er for 3 eller mere)
er for 3 email
 $P(X=0) + P(X=1) + P(X=2) + P(X=3)$

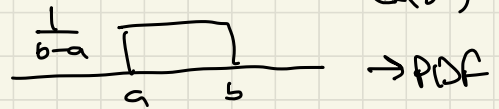
CDF



$$F_x(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x \leq b \\ 1 & x > b \end{cases}$$

$$f_x(x) = F'_x(x) = 0 \text{ for } x < a \text{ or } x > b$$

$$\text{for } a \leq x \leq b \Rightarrow f_x(x) = \frac{d(x-a)}{dx(b-a)} = \frac{1}{b-a}$$



PDF

$$f_x(u) = \lim_{\Delta \rightarrow 0} \frac{P(u \leq x \leq u+\Delta)}{\Delta}$$

$$= \lim_{\Delta \rightarrow 0} \left[\frac{F_x(u+\Delta) - F_x(u)}{\Delta} \right]$$

$$= \underline{F'_x(u)} \longrightarrow f_x(u) = F'_x(u)$$

$$\downarrow \text{PDF} = \downarrow \text{CDF in derivative}$$

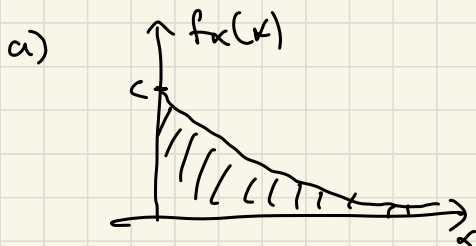
OR

$$f_x(x) = \begin{cases} c e^{-x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad \text{PDF}$$

a) CDF bul

b) CDF bul

c) $P(1 < x < 3)$ bul



$$\int_{-\infty}^{\infty} f_X(x) dx = 1 \quad \text{since}$$

$$\rightarrow \int_0^{\infty} c e^{-x} dx = -c e^{-x} \Big|_0^{\infty} = 0 - (-1) = 1$$

$c = 1 //$

b) $F_X(x) = \int_{-\infty}^x f_X(u) du = 0$ for $x < 0$

$x > 0$ then $F_X(x) = \int_0^x f_X(u) du = \int_0^x e^{-u} du = -e^{-u} \Big|_0^x = 1 - e^{-x}$

CDF $\leftarrow F_X(x) = \begin{cases} 1 - e^{-x} & x \geq 0 \\ 0 & \text{else} \end{cases} = \text{CDF}$

c) $P(1 < X < 3) = F(3) - F(1) = e^{-1} - e^{-3}$ 1.901

$\int_1^3 \underbrace{f_X(x)}_{\text{PDF}} dx = \int_1^3 e^{-x} dx = -e^{-x} \Big|_1^3 = e^{-1} - e^{-3}$ 2.901

Expected value of a Continuous Random Variable :

discrete

PMF

$\sum$

continuous

PDF

$\int$

discrete X

$$EX = \sum_k x_k P_X(x_k)$$

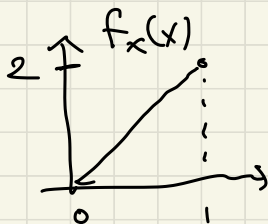
X continuous

$$EX = \int_{-\infty}^{\infty} x f_X(x) dx$$

or $f_X(x) = \begin{cases} 2x & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$

a) find the expected value of X

a) $R_x = [0, 1]$



$$EX = \int_0^1 x (2x) dx = 2 \frac{x^3}{3} \Big|_0^1 = \boxed{\frac{2}{3}}$$

Expected value of a Function of C.R.V.

$$E[g(x)] = \int_{-\infty}^{\infty} g(x) f_x(x) dx \rightarrow \text{LOTUS formula}$$

or $f_x(x) = \begin{cases} x + \frac{1}{2} & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$

find $E(x^n)$?

$$\begin{aligned} E(x^n) &= \int_0^1 x^n (x + \frac{1}{2}) dx = \int_0^1 \left(x^{n+1} + \frac{x^n}{2} \right) dx \\ &= \frac{x^{n+2}}{n+2} + \frac{x^{n+1}}{2n+2} \Big|_0^1 = \frac{1}{n+2} + \frac{1}{2n+2} = \frac{3n+4}{2n^2+6n+4} \end{aligned}$$

Variance for CRV:

$$\text{Var}(X) = E[(X - \mu_X)^2] = EX^2 - (EX)^2$$

or $f_x(x) = \begin{cases} \frac{3}{x^4} & x \geq 1 \\ 0 & \text{else} \end{cases} \quad \text{Var}(X) = ?$

$$\text{Var}(X) = \int_1^{\infty} x^2 \left(\frac{3}{x^4}\right) dx - (\mu_X)^2$$

$$= -3x^{-1} \Big|_1^{\infty} - (\mu_X)^2 = 0 - (-3) = 3$$

$$\left(\int_1^{\infty} x \left(\frac{3}{x^4}\right) dx \right)^2 = \frac{3x^{-2}}{-2} \Big|_1^{\infty} = \left(\frac{3}{2}\right)^2 = \frac{9}{4}$$

$$\text{Var}(X) = 3 - \frac{9}{4} = \left(\frac{3}{4}\right)$$

Example: Let X be a uniform $(0,1)$ random variable, and
Let $Y = e^X$

a) CDF of Y

b) PDF of Y

c) EY

$$\text{PDF}^+ \quad f_X(x) = \begin{cases} 1 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{CDF}^+ \quad F_X(x) = \begin{cases} 0 & x < 0 \\ x & 0 \leq x \leq 1 \\ 1 & x > 1 \end{cases}$$

Step 1 $\rightarrow R_Y = ?$

Step 2 $\rightarrow$ find CDF of Y $F_Y(y) = P(Y \leq y) = P(e^X \leq y)$

Step 3 $\rightarrow$ if CDF is continuous $\rightarrow f_Y(y) = \frac{dF_Y(y)}{dy}$

1-) $R_X = [0,1], Y = e^X \rightarrow R_Y = [1, e]$

$$2-) \quad F_Y(y) = \begin{cases} 0 & y < 1 \\ \ln y & 1 \leq y \leq e \\ 1 & y \geq e \end{cases}$$

3-) Y is a continuous RV

$$f_Y(y) = \frac{dF_Y(y)}{dy} = \begin{cases} 0 & \text{otherwise} \\ \frac{1}{y} & 1 \leq y < e \end{cases}$$

$$c-) EY = \int_1^e y \left(\frac{1}{y} \right) dy = y \Big|_1^e = e - 1 //$$

again remember

$$EY = E[g(x)] = \int_0^1 e^x(1) dx = e^x \Big|_0^1 = e - 1 //$$